Donthikurthi S S Bharadwaj

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Education

Vignan's Institute of Information Technology (VIIT) — Visakhapatnam, India
B.Tech in Computer Science & Engineering, CGPA: 8.5 — *2023 – 2027*

Experience

AI Backend Engineer Intern — May 2026 – June 2026
Naval Dockyard (Indian Navy), AI Department — *Visakhapatnam, India*

- Built a FastAPI endpoint (`POST /predict`) returning a predicted department and confidence score as JSON
- Implemented semantic retrieval over 70,000+ maintenance records with Sentence Transformers embeddings in PostgreSQL (pgvector), ranking departments by cosine similarity; 84.6% accuracy on a held-out test set
- Configured offline model loading; deployed the web interface on IIS and tested end to end from networked PCs

Full-Stack Developer Intern — Jun 2025 – Aug 2025
Velocitix AI — *Remote*

- Contributed to a MERN and Flask web platform, working on backend workflows and REST API integrations
- Integrated speech-to-text and LLM APIs for mock interview transcription and response evaluation
- Developed REST APIs for data flow and real-time notifications; debugged API integration issues

Projects

ProedX AI | *React, Node.js, Express, MongoDB, Flask, Docker* — Live | GitHub

- Built a end to end career assessment platform with dashboards for four user roles (students, admins, colleges, HR) using React, Node.js/Express, and MongoDB Atlas
- Built Flask AI services: Gemini API for interview questions and evaluation, Whisper speech-to-text, OpenCV/MediaPipe
- Deployed the frontend on Vercel and backend/AI services on Render, using Docker for Python services

Nexus AI | *Python, FastAPI, React, ChromaDB, Sentence Transformers* — Live | GitHub

- Built a RAG support assistant over six local Markdown docs: section-based chunking, Sentence Transformers embeddings in ChromaDB, and an LLM (OpenRouter) answering from retrieved context
- Built FastAPI endpoints (`POST /chat`, `GET /tickets`) with a keyword-based ticket vs. documentation router
- Built the React chat UI; tested pipeline stages with standalone scripts and sample questions

Converge | *React, Node.js, Express, Socket.IO, MongoDB, JWT* — Live | GitHub

- Developed a collaboration workspace for technical interviews with React, Tailwind CSS, and JWT-authenticated Express APIs, using MongoDB (Mongoose) for persistence
- Implemented Socket.IO real-time features (editing, chat, presence); debugged typing-indicator and participant issues
- Built React screens for authentication, navigation, and profile editing; investigated API authentication, environment-variable, and deployment issues across client and server

Multilingual Speech-to-Text | *React, FastAPI, faster-whisper* — Live | GitHub

- Built a React app that records microphone audio in 5-second chunks and uploads them to the backend
- Implemented a FastAPI endpoint using faster-whisper to transcribe Hindi, English, and mixed speech

Technical Skills

Languages: Python, JavaScript, C, SQL
AI Backend: FastAPI, Flask, REST APIs, LLM Integration (Gemini/OpenAI), Sentence Transformers, pgvector, Semantic Search, Retrieval Systems, Evaluation Pipelines, RAG
Databases: PostgreSQL, MongoDB, MySQL
Full Stack: (MERN) React, Node.js, Express.js, Tailwind CSS
Tools: Git, Render, Vercel, Cloudinary

Achievements

Competitive Programming: Solved 300+ problems on LeetCode; Active participant on Codeforces and CodeChef.